→ Term projects

→ Tutorial session (in 2 weeks)
(Need laptop)

Many Particles in a box — TISE (Time-independent), No particle interaction

N particles : $H = -\sum_{i=1}^{N} \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_i^2}$

$\Downarrow$

$-\sum_{i=1}^{N} \frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x_i^2} = E\psi$ $\qquad \iint\cdots\int \psi^*\psi\, dx_1 dx_2 \cdots dx_N = 1$

eg: 2 particles:

$\frac{-\hbar^2}{2m}\frac{\partial^2\psi}{\partial x_1^2} - \frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x_2^2} = E\psi$

↓

Solving for ψ: Separation of variables: $\psi(x_1, x_2) = \phi_1(x_1)\,\phi_2(x_2)$

$\Rightarrow -\frac{\hbar^2}{2m}\phi_1''\phi_2 - \frac{\hbar^2}{2m}\phi_1\phi_2'' = E\phi_1\phi_2$

$\Rightarrow \underbrace{-\frac{\hbar^2}{2m}\frac{\phi_1''}{\phi_1}}_{f(x_1)} - \underbrace{\frac{\hbar^2}{2m}\frac{\phi_2''}{\phi_2}}_{f(x_2)} = E = \text{constant} = -\lambda^2$ (λ = real)

$\therefore\ -\frac{\hbar^2}{2m}\frac{d^2\phi_1}{dx_1^2} = (E-\lambda^2)\phi_1$; $-\frac{\hbar^2}{2m}\frac{d^2\phi_1}{dx_2^2} = \lambda^2\phi_2$

BC's: 1) $\psi(0, x_2) = 0$ 2) $\psi(L, x_2) = 0 = \psi(x_1, 0) = \psi(L, x_1)$

↳ $\phi_1(0) = \phi_1(L) = 0$; $\phi_2(0) = \phi_2(L) = 0$

$\therefore\ \phi_1(x_1) = A_1 \sin\left(\frac{2m(E-\lambda^2)}{\hbar^2}x_1\right) + \cancel{\cos(\cdots)}^{\,0}$

$\phi_2(x_2) = A_2 \sin\left(\sqrt{\frac{2m\lambda^2}{\hbar^2}}\,x_2\right) + \cancel{\cos(\cdots)}^{\,0}$

$\sqrt{\frac{2m(E-\lambda^2)}{\hbar^2}}\,L = n_1\pi$ $\qquad$ $\sqrt{\frac{2m\lambda^2}{\hbar^2}}\,L = n_2\pi$

$n_1 = 1, 2, 3 \ldots$ $\qquad$ $n_2 = 1, 2, 3, \ldots$

$E - \lambda^2 = \frac{n_1^2\pi^2\hbar^2}{2mL^2}$; $\lambda^2 = \frac{n_2^2\pi^2\hbar^2}{2mL^2}$

$$\boxed{E = \frac{(n_1^2 + n_2^2)\pi^2\hbar^2}{2mL^2} \qquad n_1, n_2 = 1, 2, \ldots}$$

Any linear combination of $\psi_{n_1 n_2}$'s is also a valid solution

$\psi(x_1, x_2) = \phi(x_1)\phi(x_2) = A_1A_2 \sin\left(\frac{n_1\pi x_1}{L}\right)\sin\left(\frac{n_2\pi x_2}{L}\right)$

↳ Normalization: $A_1A_2 = \frac{2}{L}\sqrt{\frac{1}{L}}$ $\Rightarrow$

$$\boxed{\psi_{n_1,n_2}(x_1, x_2) = \sqrt{\frac{2}{L}}\sin\left(\frac{n_1\pi x_1}{L}\right)\sqrt{\frac{2}{L}}\sin\left(\frac{n_2\pi x_2}{L}\right)}$$

Q) What is the lowest energy state for a system of multiple particles (eg e⁻)?

Naive answer: $n_1 = n_2 = 1$ $\Rightarrow$ $E_{GS} = \frac{\pi^2\hbar^2}{mL^2}$

$\psi_{1,1} = \sqrt{\frac{2}{L}}\sin\left(\frac{\pi x_1}{L}\right)\sqrt{\frac{2}{L}}\sin\left(\frac{\pi x_2}{L}\right)$

This analysis is correct for bosons (integer spin)
Relativistic QM, we know that fermions (half-integer spin) are e⁻
$\psi(x_1, x_2) = -\psi(x_2, x_1)$
(note: $\psi^*\psi(x_1, x_2) = \psi^*\psi(x_2, x_1)$)

1, 1 ↑ quantum no

If we take $\psi_{GS}(x_1, x_2) = \phi_{1,1}(x_1)\ \phi_{2,1}(x_2)$
$\phi_{1,1}(x_1)\,\phi_{2,1}(x_2) = \phi_{1,1}(x_2)\,\phi_{2,1}(x_1)$
(same because $\phi_1 = \phi_2 = \phi = \sqrt{\frac{2}{L}}\sin\left(\frac{\pi x}{L}\right)$)

↳ Not true for fermions

For fermions w/o spin, we cannot have more than one particle at the lowest energy state

$\Rightarrow$ We can use Slater orbitals to ensure that linear combination of $\phi_{n_1}(x_1)$ & $\phi_{n_2}(x_2)$ are anti symmetric

We know: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = -\begin{vmatrix} c & d \\ a & b \end{vmatrix}$

S_0: $\Psi = \frac{1}{\sqrt{2}} \begin{vmatrix} \phi_1(x_1) & \phi_2(x_1) \\ \phi_1(x_2) & \phi_2(x_2) \end{vmatrix} = -\frac{1}{\sqrt{2}} \begin{vmatrix} \phi_1(x_2) & \phi_2(x_2) \\ \phi_1(x_1) & \phi_2(x_1) \end{vmatrix}$

n_1 n_2

Thus, GS for two fermions is:

$$\Psi(x_1, x_2) = \frac{1}{\sqrt{2}} \left[\phi_{n_1=1}(x_1)\, \phi_{n_2=2}(x_2) - \phi_{n_1=1}(x_2)\, \phi_{n_2=2}(x_1) \right]$$

In HW: Show

$$E = \frac{\int_0^L \int_0^L \Psi^* H \Psi \, dx_1 \, dx_2}{\int \Psi^* \Psi \, dx_1 \, dx_2}$$

w/ Ψ being antisym

For many particle systems, we can make more progress w/ DFT

Idea: Total ground state e^- density of atoms or molecules uniquely determines all electronic structure properties (i.e. energies, positions, wavefunctions)

↓

TISE: $H\Psi = E\Psi$

$$H = \underbrace{\sum_{i,e^-} -\frac{\hbar^2}{2m_e} \nabla_i^2}_{\text{electron kinetic energy}\ (1)} + \underbrace{\frac{1}{2} \sum_{i}^{N} \sum_{j \neq i} \frac{e^2}{4\pi\varepsilon_0 |r_i - r_j|}}_{\text{Coulomb repulsion}\ (2)} + \underbrace{\sum_{i}^{e^-} \sum_{j}^{\text{nuclei}} \frac{Z_\alpha e^2}{4\pi\varepsilon_0 |r_i - R_j|}}_{\text{electron-nuclei interaction}\ (3)}$$

Bohn-Heimer approximation: assumes nuclear motion is separated (since it is much heavier)

Since Ψ gives all knowable properties of system, if we know H, then we know all properties of system

If H is completed determined by total ρ (e^- density), then ρ determines all properties

Total electron density defined as:

$$n(\underline{x}) = \rho(\underline{x}) = N \iint \cdots \iint \Psi^* \Psi \, d\underline{x}_2 \, d\underline{x}_3 \cdots d\underline{x}_N$$

} Probability that any of e^-'s will be found in a box

$\therefore \int \rho(\underline{x}) \, dx\, dy\, dz = N$ and thus determines (1) and (2) of H

Kohn & Hohenberg's 1st theorem (1964)

states that $\rho(\underline{x})$ also determines term (3)

(i.e. to determine H from any potential external to the e^-)

⟹ H is completely determined by $\rho(\underline{x})$

In principle, $\rho(\underline{x})$ determines E_n & Ψ_n (not just the GS)

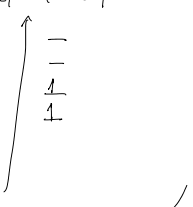
→ $F[y(x)] = \int_a^b y(x) \cdot dx$ → gives given

energy level

$= \varepsilon_1 + \varepsilon_2$

mmetric

The interpretation being one particle in ε_1 and one particle in ε_2



Simplified Pauli Exclusion principle

$(dxdydz)$

$$F[\rho] = \int f(\underline{r}, \rho(\underline{r}), \nabla\rho(\underline{r})) \cdot d\underline{r}$$

out a number
a function as

The functional derivation is

In principle, $\rho(\underline{r})$ determines E_n & ψ_n [illegible]

$F[y(x)] = \int_a^b y(x)\,dx \rightarrow$ gives given [illegible]

K & H's second theorem

States that there is a unique functional of $\rho(\underline{r})$ that, when minimized w.r.t $\rho(\underline{r})$, gives GS E and $\rho(\underline{r})$ of system: $\min_{\rho(\underline{r})} \{ E[\rho(\underline{r})] \} = E_{GS}$

Simple functional for 1 dimensional problems:

$E_{GS} = \varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_N$ $\quad N$ = total # of particles (Non-interacting fermions)

For PIB: $\varepsilon_{ni} = \frac{n_i^2 \pi^2 \hbar^2}{2mL^2}$ (ignore spin)

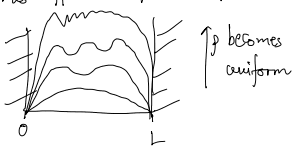
$$E = \sum_{i=1}^{N} \frac{\pi^2\hbar^2}{2mL^2} n_i^2 \quad ; \quad n_i = 1, 2, 3 \dots N$$

$$= \frac{\pi^2\hbar^2}{2mL^2} \left(\sum_{i=1}^{N} n_i^2 \right) \rightarrow \text{can be evaluated analytically}$$

$$E \approx \frac{\pi^2\hbar^2}{2mL^2} \left(\frac{1}{3}N^3 + \frac{1}{2}N^2 + \frac{1}{6}N \right) \approx \frac{\pi^2\hbar^2}{2mL^2} \frac{N^3}{3}$$

dominant for Large N

$\rho(x)$ becomes approx uniform in space



Thus if ρ = constant, $N = \int_0^L \rho(x)\,dx \approx \rho \int_0^L dx = \rho L$

$$\therefore \rho = \frac{N}{L}$$

So, $\boxed{\frac{E}{N} = \frac{\pi^2\hbar^2}{6mL^2} N^2 = \frac{\pi^2\hbar^2}{6m} \rho^2}$ $\rightarrow$ KE per particle (non-interacting fermions) ; $e_{kin}(\rho)$

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a function as
ut

The functional derivation is

$$\frac{\delta F}{\delta \rho(r)} = \frac{\partial f}{\partial \rho} - \nabla \cdot \frac{\partial f}{\partial \nabla \rho}$$

and this can be used to find the ground state when functional derivative is set to zero